

Ali Sheikholeslami

Voltage Follower, Part IV

Welcome to the 36th article in the "Circuit Intuitions" column series. As the title suggests, each article provides insights and intuitions into circuit design and analysis. These articles are aimed at undergraduate students but may serve the interests of other readers as well. If you read this article, I would appreciate your comments and feedback as well as your requests and suggestions for future articles in this series. Please e-mail me your comments at ali@ece.utoronto.ca.

In the previous three articles [1], [2], [3] in this series, we reviewed the general characteristics of a simple voltage follower (SVF), as shown in Figure 1(a), and compared them against those of a flipped voltage follower (FVF) [4], as shown in Figure 2(a), and a super-source follower (SSF) [4], [5], as shown in Figure 2(b). An SVF exhibits an infinite input resistance, a small output resistance in the order of $1/g_{me}$, and a voltage gain in the order of g_m/g_{me}, where g_m and g_{me} are the transistor's short circuit transconductance and the *effective* short circuit transconductance including its body effect. In contrast, both the FVF and SSF exhibit a significantly reduced output resistance, by a factor of $g_m r_o$, while maintaining the same voltage gain as that of an SVF (r_o is the transistor's output resistance). In this article, we highlight a major limitation that is common among all three designs, then combine the FVF and SSF designs to arrive at an improved voltage follower, known as the *class AB SSF* (*AB-SSF*) [6]. For simplicity, we neglect all the parasitic capacitances of the transistors and consider only the load capacitance (C_L). Furthermore, we assume that the capacitor C_B in the AB-SSF design is open circuit at dc and short circuit at all other frequencies.

A major limitation shared by all three designs is an asymmetry in their positive and negative slew rates. To illustrate this, we resort to the SVF simulation results shown in Figure 1(b). We apply an input voltage waveform that suddenly falls by 0.4 V and observe that the output voltage cannot quickly follow this change. This is simply because the voltage across a capacitor cannot change suddenly; it can change only at a rate dictated by the current supplied to it. In other words

$$\frac{dv_{out}}{dt} = \frac{i_L}{C_L}.$$

At the falling edge, when the gate-to-source voltage of M_1, v_{GS1}, falls momentarily below the threshold voltage of M_1, M_1 is turned off, and $i_L = -I_B$. Substituting this into the previous equation yields the maximum rate of change at the output node in the negative direction, defined as the *negative slew rate* (SR^-), and its absolute value is given by $SR^- = I_B/C_L$.

On the other hand, the positive slew rate, i.e., the maximum rate of change at the output node in the positive direction, is not limited by the bias current. Indeed, the current that charges the output capacitor is directly influenced by the transistor size and its gate-to-source voltage. When the input suddenly rises from 0.6 to 1 V, it creates a v_{GS} well above its bias value, causing the transistor

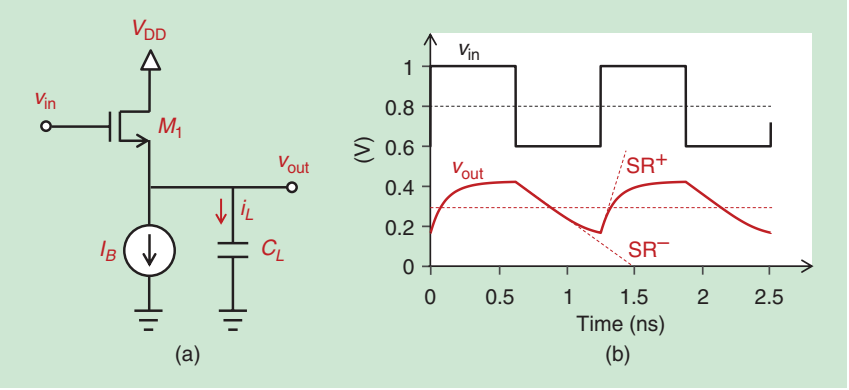


FIGURE 1: (a) A schematic of a simple voltage follower (SVF) using an NMOS transistor biased with a dc current source I_B. (b) An input waveform and its corresponding simulated output waveform reveal an asymmetry in the negative and the positive slew rates, SR^- and SR^+, respectively.

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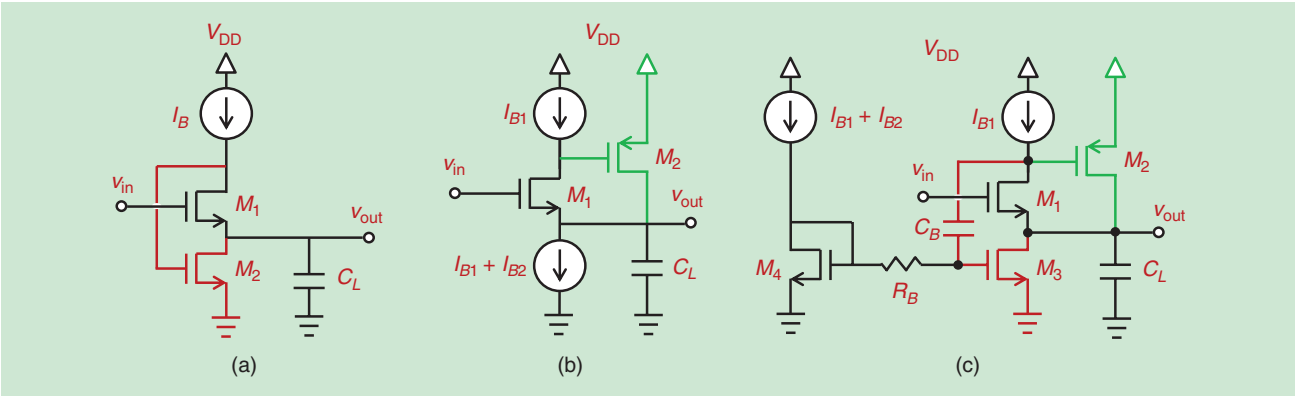


FIGURE 2: Transistor-level of schematics of improved voltage followers. (a) Flipped voltage follower (FVF). (b) Super-source follower (SSF). (c) Class AB SSF (AB-SSF), a combined FVF and SSF.

to source a current much larger than its bias current to the capacitor. In fact, with the design parameters shown in Table 1, transistor M_1 can source a current that is five to six times larger than its bias current, resulting in an SR^+ that is five to six times that of SR^-. This ratio can be observed in the slopes depicted in Figure 1(b).

This simple example illustrates how the design of the SVF has an asymmetric slew rate in response to a sudden rise/fall of an input waveform. Interestingly, the FVF exhibits an asymmetry in the opposite direction. This is simply because we have "flipped" the direction of the current source in the FVF design, thus limiting the rate of positive change for the output node voltage. In other words, for the FVF, we have $SR^+ < SR^-$. How about the SSF? The readers are encouraged to pause here and explain why in the SSF the asymmetry in the slew rate is similar to that of the SVF. In other words, $SR^+ > SR^-$.

We wish to come up with a design that removes this asymmetry and presents the larger slew rate for both the rising and falling edges of the waveform. One way to improve the slew rate is to simply increase bias current(s), but this leads to an increase in power consumption, which we ideally want to avoid. Another way is to use the complement version of the circuits, swapping NMOS and PMOS devices. This solution simply swaps the values of SR^+ and SR^- but does not solve the asymmetry problem.

TABLE 1. NMOS AND PMOS TRANSISTOR PARAMETERS USED IN SIMULATIONS OF SVF, FVF, SSF, AND AB-SSF.

PARAMETER	M_1 (IN ALL DESIGNS) AND M_2 IN FVF	M_2 (SSF AND AB-SSF)	M_3 (AB-SSF)
Type	NMOS	PMOS	NMOS
W/L	400 nm/45 nm	1,150 nm/45 nm	800 nm/45 nm
$\|V_T\|$	0.43 V	0.27 V	0.43 V
g_m	0.5 mS	0.5 mS	0.79 mS
g_{mb}	0.15 mS	0.15 mS	0.23 mS
r_o	10.5 kΩ	9.3 kΩ	6.8 kΩ

The solution presented in Figure 2(c) combines the FVF and the SSF designs into an SSF with class AB operation. We will show that this solution maintains all the advantages of the FVF and the SSF (with respect to the SVF) while offering symmetric slew rates, not limited by the size of the bias currents. In the remainder of this article, we first show intuitively how the AB-SSF maintains the FVF and the SSF advantages (such as having a low output resistance) and then demonstrate its symmetric slew-rate behavior, and finally, describe its class AB operation.

We have color-coded the transistors in Figure 2 to reveal how the circuit of Figure 2(c) is indeed a combination of the FVF and SSF along with an RC circuit formed by a large resistor R_B and an ac coupling capacitor C_B. At the center of this circuit lies the source follower transistor M_1 with a constant dc current I_{B1} and, hence, a constant v_{GS1}, performing the task of "voltage following." The other two transistors, M_2 and M_3, perform the task of current delivery to the capacitive load as we now describe. Under dc conditions, i.e., when the input is at a constant dc level, there is no current flowing through the load simply because the capacitor acts as an open circuit at dc. In this case, the dc currents through M_1 and M_2 are I_{B1} and I_{B2}, respectively, and the current through M_3 is $I_{B1} + I_{B2}$. Now, let us apply a small-signal input voltage v_{in} to this circuit and derive an expression for its output voltage.

We calculate the output voltage using $v_{out} = i_{sc} \times R_{out}$, where i_{sc} represents the short circuit current from the output node to ground, and R_{out} represents the equivalent resistance looking into the output

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node [7]. Figure 3(a) shows a small-signal model of the AB-SSF where we have grounded all the dc supplies and further grounded the output node to measure its short circuit current. In doing so, we have also assumed R_B to be very large and C_B to act like a short circuit in the frequency range of interest. According to this circuit, the short circuit current is $i_{sc} = i_1 + i_2 + i_3$, where i_1, i_2, and i_3 are the small-signal currents by each of M_1, M_2, and M_3, respectively, as shown. Given that i_1 originates from an open circuit (the gates of M_2 and M_3 have infinite resistance), we can conclude that $i_1 = 0$. Nevertheless, M_1 amplifies v_{in} to a much larger voltage with opposite polarity at its drain. This voltage, which can be found easily (as in [2], [3]) to be in the order of $v_{d1} = -g_{m1} r_{o1} v_{in}$, is applied simultaneously to the gates of M_2 and M_3, creating $i_2 = g_{m2} g_{m1} r_{o1} v_{in}$ and $i_3 = g_{m3} g_{m1} r_{o1} v_{in}$. Therefore, we can write

$$i_{sc} = (g_{m2} + g_{m3}) g_{m1} r_{o1} v_{in}$$

It is interesting to note that this short circuit current is the sum of two short circuit currents: one from the SSF and one from the FVF as we found them in [2], [3]. In fact, if we assume all transistors to have the same short circuit transconductances, this equation states that the short circuit current in the AB-SSF is twice as large as those in the FVF and the SSF.

The reader is encouraged to use the equivalent circuit in Figure 3(b) to show that the output resistance of the AB-SSF is also equivalent to the parallel combination of the output resistances of those of the FVF and the SSF as we found them in [2], [3]. It is easy to show that

$$R_{out} = \frac{1}{(g_{m2} + g_{m3})(g_{me1} r_{o1})}$$

where we have assumed $g_m r_o \gg 1$ in all transistors. This expression indicates that the AB-SSF reduces the output resistance to half of those of the FVF and the SSF if we assume $g_{m2} = g_{m3}$. Finally, we can write

$$v_{out} = i_{sc} \times R_{out} = g_{m1} \frac{v_{in}}{g_{me1}}$$

which is the same expression we found for the output voltages of the FVF and the SSF.

To recap, so far, we have shown that the AB-SSF provides the same low-frequency voltage gain as those of the FVF and the SSF but does so with twice the short circuit current and half the output resistance. Let us now observe the AB-SSF output behavior in response to a large signal input and compare this behavior against those of the FVF and the SSF. We apply an input voltage with a rectangular waveform at 800 MHz, as shown in Figure 4(a), and plot the simulated output waveforms (ac responses only) for each of the FVF (in red), the SSF (in green), and the AB-SSF (in blue) designs in Figure 4(b). The parameters used for these simulations are summarized in Table 1. We first observe that all three designs exhibit a similar swing at the output node. Second, we observe that while the output voltages of the FVF and the SSF exhibit limited slew rates at their rising and falling edges, respectively, the output of the AB-SSF does not exhibit either limitation. Now, let us look back at the FVF, SSF, and AB-SSF

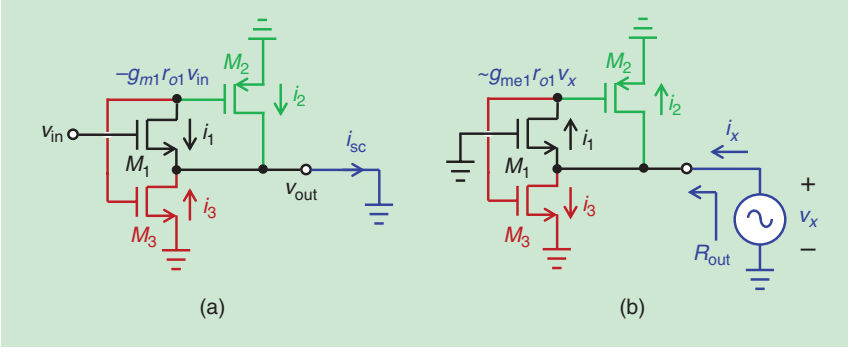


FIGURE 3: A small-signal equivalent circuit for the AB-SSF (a) to find the short circuit current at the output node and (b) to find the output resistance.

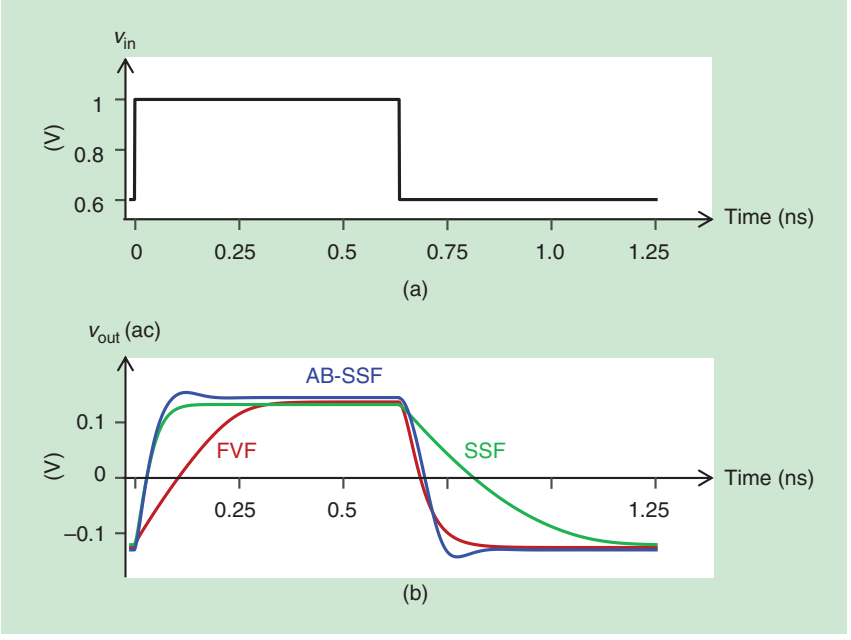


FIGURE 4: (a) A sketch of a large-signal input voltage waveform at 800 MHz. (b) Simulated ac output voltage waveforms for the three voltage followers. The curves in red and green highlight the limited slew rates at the rising and falling edges of the FVF and the SSF, respectively.

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circuits in Figure 2 to determine if these results make intuitive sense.

A rising input will force the output node to rise in all three designs. However, the rate of rise at the output node, which we called the *positive slew rate*, or SR^+, is constrained to I_B/C_L for the FVF, irrespective of the size of the input rise. This is because the maximum current we can supply to C_L is limited to I_B in the FVF. Interestingly, this same constraint does not exist in either the SSF or the AB-SSF because, in these designs, the current supplied to the load during the rise is provided by the PMOS transistor M_2, which can supply currents well above its bias current level.

Similarly, a falling input will force the output node to fall in all three designs. However, the rate of fall at the output, SR^-, is constrained to I_{B2}/C_L in the SSF but not in the FVF and the AB-SSF, where M_2 in FVF and M_3 in AB-SSF can sink currents well above their bias current levels. With $I_{B1} = I_{B2} = 50\,\mu$A in our designs, the SR^+ of the FVF is similar to the SR^- of the SSF, as can be observed in Figure 4(b). The AB-SSF enjoys the best of both designs.

To further compare the slew behavior of the three designs, we consider a 3-GHz sinusoidal input voltage waveform with 0.4 Vpp, as shown in Figure 5(a), and plot its corresponding simulated output waveforms in the three designs. Given that $v_{in}(t) = 0.2\sin(2\pi ft)$, with $f = 3$ GHz, and assuming a similar waveform at the output, say $v_{out}(t) = 0.14\sin(2\pi ft + \phi_0)$, the expected maximum rate of change of the output can be written as $\max(dv_{out}/dt) = 0.28\pi f$. When this rate exceeds the slew-rate limit of the design, the output can no longer track the input, and its rate will be limited to the slew rate. This behavior can be observed in Figure 5(b) where the FVF and the SSF show slew behaviors during the output rise and fall, respectively, whereas the AB-SSF is not affected.

Let us now describe the class AB operation of the AB-SSF circuit through simulation results. We apply a 3-GHz sinusoidal voltage waveform with a 0.4-Vpp swing to the input node and observe its output voltage, its output current, and its transistor currents. Figure 6(a)

(continued on p. 19)

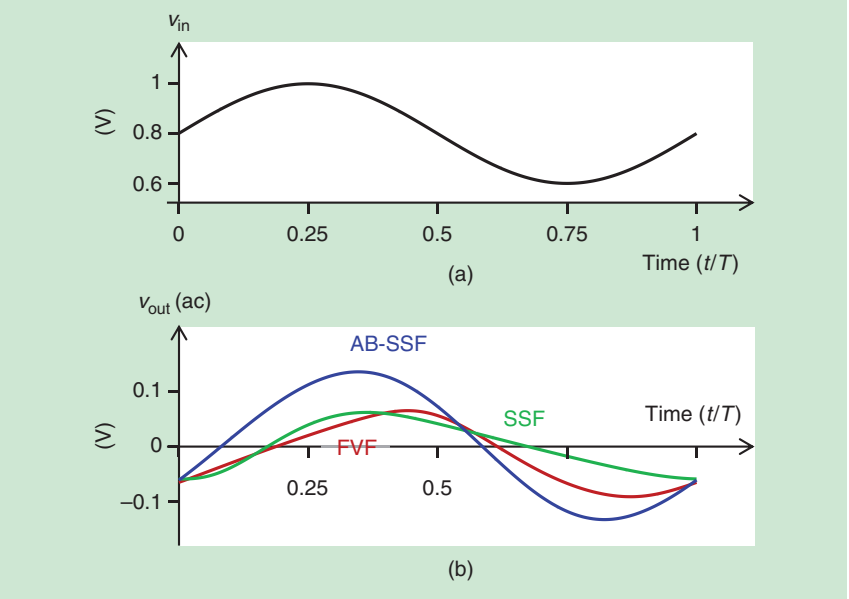


FIGURE 5: The impact of the slew-rate limit on a sinusoidal output waveform at 3 GHz: (a) A sketch of the input voltage waveform. (b) Simulated ac output voltage waveforms for the three voltage followers: FVF in red, SSF in green, and AB-SSF in blue.

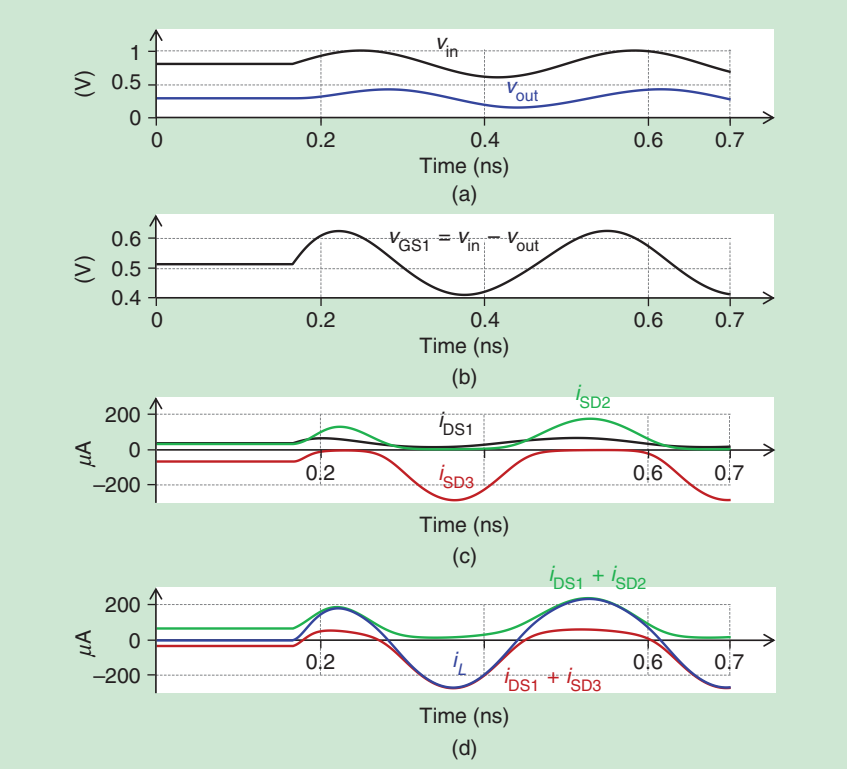


FIGURE 6: Voltage and current waveforms of the AB-SSF circuit. (a) Input and output voltage waveforms, (b) the gate-to-source voltage of M_1. (c) The current waveforms of M_1 (in black), M_2 (in green), and M_3 (in red). (d) The current contributions of M_1 and M_2 (in green) and M_1 and M_3 (in red) to the total load current (in blue).

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Appendix: Related Articles

[A1] C. Auth and A. Shankar, "Evolution of transistors: Humble beginnings to the ubiquitous present," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 20–28, Summer 2023, doi: 10.1109/MSSC.2023.3281538.

[A2] L. W. Nagel and C. C. McAndrew, "The evolution of transistor models: Always trying to keep up with the evolution of transistors," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 29–35, Summer 2023, doi: 10.1109/MSSC.2023.3284805.

[A3] M. Ikeda, "Reflections on progress in digital circuits: From basics to untangle past trials," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 36–42, Summer 2023, doi: 10.1109/MSSC.2023.3290099.

[A4] Y. Chae, C. M. Lopez, K. A. A. Makinwa, M. Ortmanns, and W. Sansen, "A glimpse of the history of analog ICs," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 43–52, Summer 2023, doi: 10.1109/MSSC.2023.3282557.

[A5] T.-Y. J. Chang, F. Hamzaoglu, Y. Li, K. Sohn, Y. Wang, and J. Wuu, "A trip down memory lane: Reflections on the progress in memories," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 53–61, Summer 2023, doi: 10.1109/MSSC.2023.3287169.

[A6] B. Razavi, "75 years of RF design: Highlights and paradigm shifts," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 62–79, Summer 2023, doi: 10.1109/MSSC.2023.3280435.

[A7] N. Collaert, "Advancements in IC technologies: A look toward the future," *IEEE Solid-State Circuits Mag.*, vol. 15, no. 3, pp. 80–86, Summer 2023, doi: 10.1109/MSSC.2023.3280433.

CIRCUIT INTUITIONS *(continued from p. 9)*

shows that the output voltage follows the input voltage, albeit with a voltage drop v_{GS1}, shown in Figure 6(b), that can change from 0.4 V to 0.6 V. Given that the threshold voltage of $M1$ is $V_{T1} \approx 0.4$ V according to Table 1, when v_{GS1} is close to 0.4 V, $M1$ is expected to turn off and not conduct any current. This is indeed consistent with the current waveform of i_{DS1} shown in black in Figure 6(c). Interestingly, each of $M2$ and $M3$ will also experience time intervals during which they contribute zero current to the output load; i_{SD2} in green is near zero in the (0.3-ns, 0.4-ns) time interval, and i_{SD3} in red is near zero in the (0.5-ns, 0.6-ns) time interval.

Given that the total load current i_L can be written as the sum of the three transistor currents, i.e., $i_{DS1} + i_{SD2} + i_{SD3}$, we can see how the three transistors cooperate to provide the total current to the load. The notation i_{SD2}, for example, represents the total current from source to drain in $M2$. When the maximum positive current is being delivered to the load, transistors $M1$ and $M2$ join forces to supply the required current to the load, while transistor $M3$ is off contributing zero current to the load. On the other hand, the maximum negative current is delivered to the load by $M3$ alone as both $M1$ and $M2$ conduct almost zero currents at this time. There is an interval, however, around $t = 0.45$ ns, during which all transistors are on and contribute currents to the load. This happens around the time when the input signal deviates only slightly from its bias value and all transistors conduct current to perform what is referred to as *class A operation*. At other times, only one or two transistors among the three source/sink current to the load, and the other one or two transistors are off. This is referred to as a *class AB operation* when each transistor conducts currents only for a portion (more than half) of the period and stops conducting current for the remaining portion. By superimposing $i_{DS1} + i_{SD2}$ and $i_{DS1} + i_{SD3}$ curves on top of the load current i_L curve, Figure 6(d) clearly shows how i_L is constructed by $i_{DS1} + i_{SD2}$ around its peak (when $i_{SD3} \approx 0$), by the $i_{DS1} + i_{SD3}$ curve around its trough (when $i_{SD2} \approx 0$), and by $i_{DS1} + i_{SD2} + i_{SD3}$ in other parts where all three transistors conduct current.

In summary, we have shown that the SVF, the FVF, the SSF, and the AB-SSF all offer the same voltage gain of around 0.7 V/V. The main advantage of the FVF, SSF, and AB-SSF over the SVF is their reduced output resistance, which is a major improvement in the design of a voltage follower. Among the four designs, the AB-SSF exhibits the lowest output resistance and a much better symmetric slew behavior in its output voltage thanks to its class AB operation.

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